DATA ANALYTICS LEVEL III

Day 4: Working with Data in Excel — Techniques & Industry Practice

Student Activities

How to Use This Document

This is your activity workbook for today's 4-hour meeting. You will clean real messy data and practice smarter averages. Use Excel FORMULAS where asked (your instructor will check the formula bar). Activities 1 and 2 use Day04_Dataset.xlsx.

Activity	What You Submit	Time
Activity 1: Clean & Standardize a Messy Dataset	Completed Messy Orders sheet	70 min
Activity 2: Weighted Average Practice	Completed Weighted Average sheet	40 min
Activity 3: Spot the Misleading Average	A short document	30 min
Learning Journal Entry #3	Your journal reflection	20 min

Activity 1 — Clean & Standardize a Messy Dataset

Goal: Turn a messy, inconsistent dataset into clean, analysis-ready data.

Open Day04_Dataset.xlsx and go to the Messy Orders sheet (24 orders). The City column is spelled many different ways and the prices are unformatted.

Step 1 — Standardize the City column (Find & Replace, Ctrl+H)

- Replace all the variants so every city is one of exactly four values: Manila, Cebu, Davao, Quezon City
- **Examples:** “MLA”, “manila”, and “Manila ” (trailing space) all become “Manila”; “QC” and “quezon city” become “Quezon City”
- **Tip:** also fix any stray trailing spaces (replace “Manila ” with “Manila”)

Step 2 — Build and format the Total column

- In the Total column (E), enter =C4*D4 and fill down to row 27
- Format Unit Price and Total as Currency (₱ with a thousands separator)

Step 3 — Count and total (cleanup area, use formulas)

- =COUNTIF the City column for each of the four cities
- Grand Total of the Total column (=SUM); Biggest single order (=MAX)

Step 4 — Sort and filter

1. Sort the whole table by Total, largest first
2. Turn on Filter and show only Manila orders; put their total (=SUM of the visible Manila Totals) in the cleanup area

Step 5 — Make a chart (Total Sales by City)

A cleaned dataset is ready to visualize. Build a small summary and turn it into a chart:

3. In a spare area, list the four cities and, next to each, its total sales using =SUMIF(City range, city, Total range)
4. Select that little summary table (cities + totals)
5. Insert tab → Insert Column Chart → Clustered Column
6. Give the chart a clear title: "Total Sales by City"

- **Then answer on the sheet:** which city has the highest total sales? Is it the same city that has the most orders?

Challenge: add a Data Validation dropdown (Manila/Cebu/Davao/Quezon City) to the City column so future typos are impossible.

Save as: Day04_Activity1_[YourName].xlsx

Activity 2 — Weighted Average Practice

Goal: Compute weighted averages and see why they beat the naive average.

Go to the Weighted Average sheet.

Table A — Product Star Ratings

- In the "Stars × Reviews" column, multiply each star value by its number of reviews (a formula)
- Compute Total reviews (=SUM) and the Sum of (Rating × Reviews) (=SUM)
- Weighted average rating = Sum of (Rating × Reviews) ÷ Total reviews
- Also compute the naive average = (5+4+3+2+1)/5, and answer: which is honest, and why?
- **Challenge:** get the weighted average in ONE formula with =SUMPRODUCT

Table B — Course Grade

- Compute each component's Weight × Score, then the Weighted grade (=SUM of those)
- Compute the simple average of the four scores, and compare it to the weighted grade

Save as: Day04_Activity2_Weighted_[YourName].xlsx

Activity 3 — Spot the Misleading Average

Goal: Apply professional judgment — recognize when an “average” is misleading.

For EACH scenario below, in 2–3 sentences: say why the reported average is misleading, and what you would report or recommend instead.

Scenario 1 — The Product Rating

A seller advertises “Average rating: 3.0 stars” by averaging the five star labels (5+4+3+2+1)/5. But 70 of the 100 reviews are 4 or 5 stars. Is 3.0 fair?

Scenario 2 — The Team Salary

A company says its “average salary is ₱95,000.” Four staff earn ₱28,000–₱35,000 and the owner earns ₱400,000. Is the average a fair picture of what a typical employee earns?

Scenario 3 — The Delivery Promise

A restaurant advertises “average delivery: 30 minutes.” Most orders arrive in 25 minutes, but a few faraway orders take 90+ minutes. What is the average hiding, and what should they tell customers?

Submit as: Day04_Activity3_Averages_[YourName].docx

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Learning Journal Entry #3

Goal: Reflect on data technique and professional judgment.

1. Why is cleaning data before analysis so important? Give one example from today.

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2. In your own words, when should you use a weighted average instead of a plain one?

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3. Which technique (Find & Replace, Data Validation, formatting, sort/filter) will help you most, and why?

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4. My confidence with Excel today (1 = lost, 5 = very confident), and why:

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Submit as: Day04_Journal3_[YourName].docx

Submission Checklist

- **[ ] Activity 1:** Cleaned Messy Orders sheet (standardized cities, Total column, COUNTIF, sort/filter)
- **[ ] Activity 2:** Weighted Average sheet (both tables)
- **[ ] Activity 3:** Misleading-average critique (3 scenarios)
- [ ] Learning Journal Entry #3